

Review

Ras, PI3-kinase and mTOR signaling in cardiac hypertrophy

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Abstract

Cardiac hypertrophy involves increased mass (growth) of the heart and a cardinal feature of this condition is increased rates of protein synthesis. Several signaling pathways have been implicated in cardiac hypertrophy including the phosphatidylinositol 3-kinase (PI3K) and Ras/Raf/MEK/Erk pathways. PI3K lies upstream of the mammalian target of rapamycin (mTOR), an important positive regulator of protein synthesis and cell growth. However, recent data suggest that, in response to certain hypertrophic agents, signaling via Ras and MEK/Erk, as well as mTOR, is required for activation of protein synthesis, indicating new connections between these key signaling pathways.

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Cardiac hypertrophy involves increased heart size due to increased cardiomyocyte size. Initially, it is an adaptive response to increased workload or to defects in the efficiency of the contractile machinery. However, in the longer term it contributes to the development of heart failure and sudden death.

Increased protein synthesis is a key feature of cardiac hypertrophy and likely underlies the increased cell and organ size observed under this condition [25]. Several other features are also seen, including changes in gene expression and reorganisation of the contractile machinery, and it should be borne in mind that the protein accumulation that is characteristic of hypertrophy may reflect effects on protein breakdown as well as protein synthesis. Elucidating the signaling connections underlying cardiac hypertrophy is important for both our fundamental understanding of the process and developing potential therapeutic strategies for this condition, which is a major risk factor for cardiac failure. This is a vigorous research area and several signaling pathways have been implicated in hypertrophic responses in cardiomyocytes [19,53,83]. Much work has focused on the control of transcription in cardiomyocytes, especially neonatal cells. This article will focus primarily on the regulation of the protein synthetic machinery and principally on the roles of the Erk

(classical MAP kinase; Ras/Raf/MEK/Erk) and phosphatidylinositol 3-kinase (PI3K) pathways in cardiac hypertrophy, and their interactions with the mTOR pathway that plays a central role in regulating mRNA translation and cell growth.

1. Ras and Erk signaling

Treatment of cardiomyocytes with a variety of pro-hypertrophic agents increases the proportion of Ras in its active GTP-liganded form [84]. Among the agents which do this are the G_q/G_{11} linked agonists angiotensin II (Ang II), endothelin-1 (ET-1) and phenylephrine (PE), each of which elicits hypertrophic responses [53]. Studies using transgenic mice have revealed a key role for G_q/G_{11} in the development of cardiac hypertrophy in response, e.g., to overload [1,105] (see also Ref. [66] for a review). The mechanisms by which G-protein-coupled receptor (GPCR) agonists activate Ras is not fully understood. This likely involves signaling from the $G_{\beta\gamma}$ subunits through tyrosine kinases (reviewed in Ref. [84]). Ras-GTP activates several effector proteins including Raf (which lies upstream of the Erk/MAP kinase pathway), PI3K (upstream, e.g., of protein kinase B (PKB, also known as Akt)) and the Ral GDP-dissociation stimulator (Ral-GDS) (see Fig. 1A). Ras has also been reported to activate the c-Jun N-terminal kinases (JNKs) in cardiomyocytes [61].

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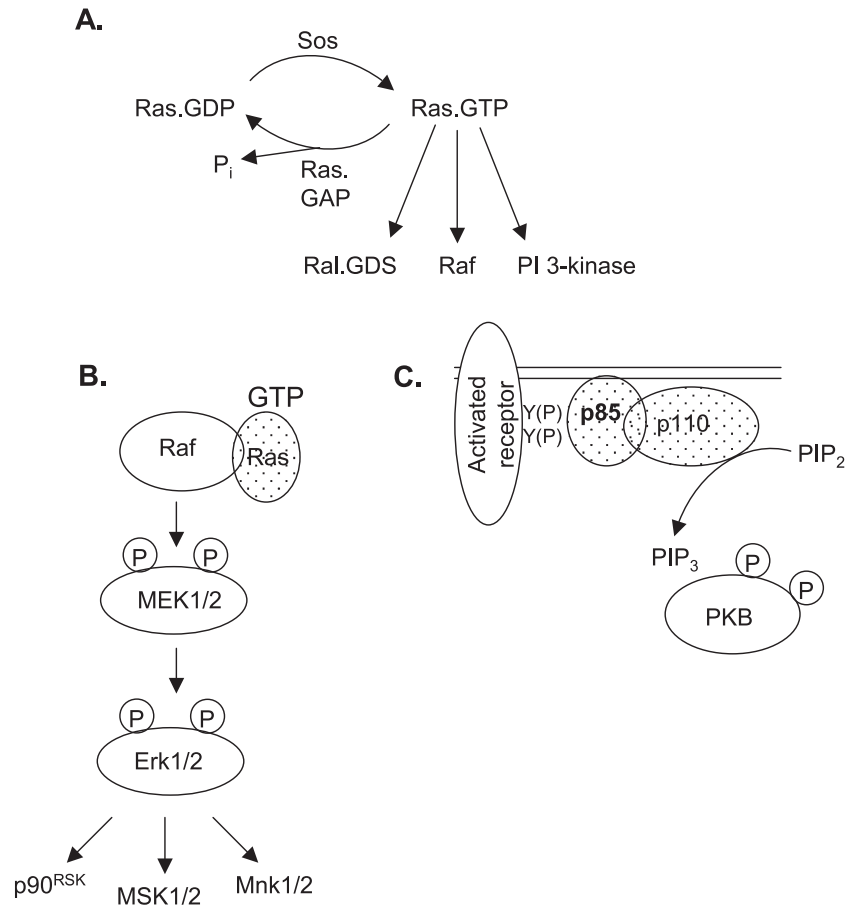


Fig. 1. Signalling pathways. (A) Ras is converted to its active GTP-liganded state by guanine nucleotide exchange factors such as Sos. Effectors of activated Ras include Ral.GDS, Raf and PI3K, which couple to downstream signaling pathways. Hydrolysis of the bound GTP (facilitated by a GAP) converts Ras to an inactive state. (B) Ras-GTP binds to and activates Raf, which phosphorylates and activates the dual-specificity kinases MEK1/2, which in turn phosphorylates and activates Erk1/2. (C) Typically, the p85 subunit of PI3K binds to phosphotyrosine residues within an activated transmembrane receptor. The catalytic p110 subunit is thereby recruited to the membrane where it catalyses phosphorylation of PIP₂ to PIP₃. PIP₃ interacts with PKB/Akt, leading to its phosphorylation and activation.

The potential importance of Ras signaling in cardiac hypertrophy is underlined by the observation that transfection or microinjection of cardiomyocytes with an activated Ras mutant (V12Ras) leads to changes in gene expression similar to those seen in hypertrophy [91,92]. Furthermore, Ras causes myofibrillar changes and other effects characteristic of hypertrophy [20,92] (for review, see Ref. [84]). Perhaps most significantly, targeted overexpression of V12Ras in mouse heart caused ventricular hypertrophy and cardiac failure [30]. Recent data suggest that overexpression of V12 Ras itself may cause general changes in gene expression, while use of mutants that preferentially activate Raf or Ral-GDS induces activation of ANF transcription (this being a gene whose expression is increased in hypertrophy). Conversely, haploinsufficiency of Grb2, an adaptor protein involved in Ras activation, impairs development of hypertrophy in response to pressure overload [110]. For a more detailed discussion of the role of Ras in cardiac hypertrophy see Ref. [84].

A key question is how active Ras promotes hypertrophic responses, given that several effectors exist for it. A variety of

studies suggest roles for Src family tyrosine kinases [39], Ral-GDS [36] or the Raf/MEK/Erk cascade [4–6,10,13,20,22,92,93]. For a detailed discussion of the roles of other MAP kinases hypertrophic responses see Ref. [19].

Recent data suggest Ras plays a key role in the activation of protein synthesis by PE, as demonstrated using the dominant interfering N17Ras mutant or inhibitors of Ras isoprenylation [99]. Furthermore, as described below, the Raf/MEK/Erk pathway appears to mediate the activation of protein synthesis by agents such as PE in cardiomyocytes. We shall therefore now take a closer look at the Raf/MEK/Erk pathway.

2. Raf/MEK/Erk signaling

This pathway involves a module of three protein kinases, Erk, MEK and Raf. Erk exists in two isoforms in mammalian cells, Erk1 and Erk2 (also called p44 and p42 MAP kinase, respectively, Fig. 1B). Phosphorylation of Erk1/2 within the Thr–Glu–Tyr (TEY) motif results in full activation, allow-

ing them to phosphorylate downstream substrates, including several other protein kinases. Selective low molecular weight inhibitors of MEK1/2 (the kinases that phosphorylate Erk1/2) have proved very useful in dissecting the roles of the Erks in cellular regulation [16]. Activation of Erk can be reversed by dual-specificity protein phosphatases [37], e.g., MAP kinase phosphatase (MKP)-3 [11,55]. MEK1/2 are also activated by phosphorylation, catalysed by members of the Raf group of kinases, and Raf is activated by binding Ras-GTP.

Cell stretch and GPCR agonists activate this pathway in neonatal or adult cardiomyocytes [4,5,13,57,70,98,99,108], while Erk1/2 are also activated by pressure overload (aortic banding) [62,86]. Using neonatal myocytes, antisense and pharmacological approaches have provided evidence that this pathway plays positive roles in hypertrophic responses. Xiao et al. [106] have provided evidence that the MEK/Erk pathway mediates hypertrophic effects of α_1 -adrenergic agonists in adult cardiomyocytes. A major end-point of their study was the activation of protein synthesis, and their data agree with the conclusion [99] that Ras/Erk signaling plays a critical role here. However, studies have provided differing, or even contradictory, data [12,93–95,109], one even suggesting that Erk signaling prevents hypertrophy [81]. There are several possible sources of discrepancies between data for the roles of different signaling pathways in hypertrophic responses: besides the usual difficulty of interpreting information obtained using different approaches, other potential problems include use of adult vs. neonatal cardiomyocytes (most studies used the latter) and the fact that various studies focus on different end-points: changes in specific gene expression, sarcomeric organisation or activation of protein synthesis/protein accumulation. Signaling connections may indeed differ between adult and neonatal cells—for example, while many studies have reported robust activation of the JNKs by a variety of pro-hypertrophic stimuli in neonatal cells (see Ref. [19]), we observe no such activation by PE or ET-1 in adult cells [98].

Transgenic mouse models provide an important resource for investigations in this area. Recent studies show that cardiac-specific expression of an activated form of MEK1 leads to development of concentric hypertrophy involving thickened septal and left ventricular walls [9]. Thus, activation of MEK/Erk signaling appears to be sufficient to induce hypertrophy in vivo. Interestingly, overexpression of Ras gave rise to a different phenotype, characterised by pathological ventricular remodelling [30]. This may reflect its ability to activate additional signaling pathways. Work using further transgenic animal models is essential to gain a fuller understanding of the roles of Erk signaling in normal myocardium and hypertrophic responses.

3. PI3K

PI3K plays an important role in controlling organ size in *Drosophila* and in mouse heart [41,42]. This makes it a

candidate for involvement in cardiac hypertrophy. PI3K phosphorylates phosphoinositides at the 3 position. The major physiological product is phosphatidylinositol 3,4,5-trisphosphate (PIP₃) for which an important downstream target is PKB/Akt, which is activated by phosphorylation in response to agents that stimulate PI3K [71,72]. Binding of PIP₃ to the pleckstrin-homology domain of PKB recruits it to the plasma membrane, where it is phosphorylated (within the so-called T-loop) by the kinase PDK1. Additional activation at a C-terminal serine is required for full activation of PKB. Several substrates for PKB are known and PI3K and/or PKB signaling is thought to play important roles in diverse processes including cell growth and survival, as well as anabolic effects of insulin [71,97].

Targeted overexpression of constitutively active PI3K in the heart resulted in increased organ size while expression of a dominant-negative mutant decreased it [78]. The sizes of the liver and lungs did not change. Interestingly, the increased heart size was associated with a similar increase in myocyte size, indicating true hypertrophy. To address whether PKB/Akt was the relevant effector of PI3K, these workers employed mice expressing constitutively active or kinase-deficient PKB mutants specifically in the heart [79]. Constitutively active PKB increased heart and myocyte size suggesting that PKB is a key mediator of these growth effects. Two other studies have also shown that transgenic expression of PKB in the heart leads to increased size of the heart and of cardiomyocytes [15,47] (although it should be noted that other phenotypic effects were also observed). Consistent with this hypertrophic effect of active PKB, Shioi et al. [78] observed that expression of kinase-deficient PKB partially reversed the growth-promoting effect of constitutively active PI3K, while the active PKB mutant promoted heart growth to the same extent in mice expressing the dominant-negative PI3K as those expressing wildtype PI3K. This suggests that activation of PKB circumvents the growth-suppressive effects of dominant interfering PI3K. The lipid phosphatase PTEN acts to oppose the effects of PI3-kinase by dephosphorylating PIP₃. Expression of a catalytically inactive PTEN mutant in the heart also causes hypertrophy, presumably by virtue of its ability to activate PKB signalling [76]. It is important to point out here that the PI3K pathway has been implicated, on the basis of elegant genetic studies, in controlling organ and animal size in fruit-flies and nematodes (for recent reviews, see Refs. [41,46]). Control of cell and organ size is likely linked to regulation of protein synthesis and we shall return to this point. PTEN.

In mice expressing dominant-negative PI3K in the heart [50], exercise-induced ('physiological') but not overload-induced ('pathological') hypertrophy was impaired, suggesting that latter is mediated by other pathways. The data suggest PI3K plays a role in physiological hypertrophic responses and it is thus relevant to note that exercise led to activation of PKB (Akt) but not Erk or p38 MAP kinase, which may be involved in other types of hypertrophic response.

IGF1 is a candidate for mediating the ‘physiological’ hypertrophic effect. Its levels are increased by exercise [38,40,107]. Enhanced expression of the local form of IGF1 led to a hypertrophic phenotype [17]: in contrast, overexpression of IGF1 in the myocardium causes increased numbers of cardiomyocytes, rather than hypertrophy [64,65].

Taken together, these data suggest that PI3K and PKB are potential mediators of certain hypertrophic responses in heart. This probably involves activation of protein synthesis, leading one to ask what the relevant targets of this signaling module might be. There have recently been major advances in this area, as discussed below. However, I will first describe in brief mechanisms by which the

activity of the translational machinery is controlled in mammalian cells.

4. Overview of regulation of mRNA translation

The process of mRNA translation is conventionally divided into three main stages: initiation, elongation and termination. During initiation, the small ribosomal (40S) subunit is first recruited to the 5'-end of the mRNA, via the latter's cap structure which contains a 7-methylguanosine triphosphate moiety linked via a 5'–5' phosphodiester bond to the mRNA (Fig. 2A). Initiation requires a set of proteins termed eukaryotic initiation factors (eIF's). eIF4E binds to

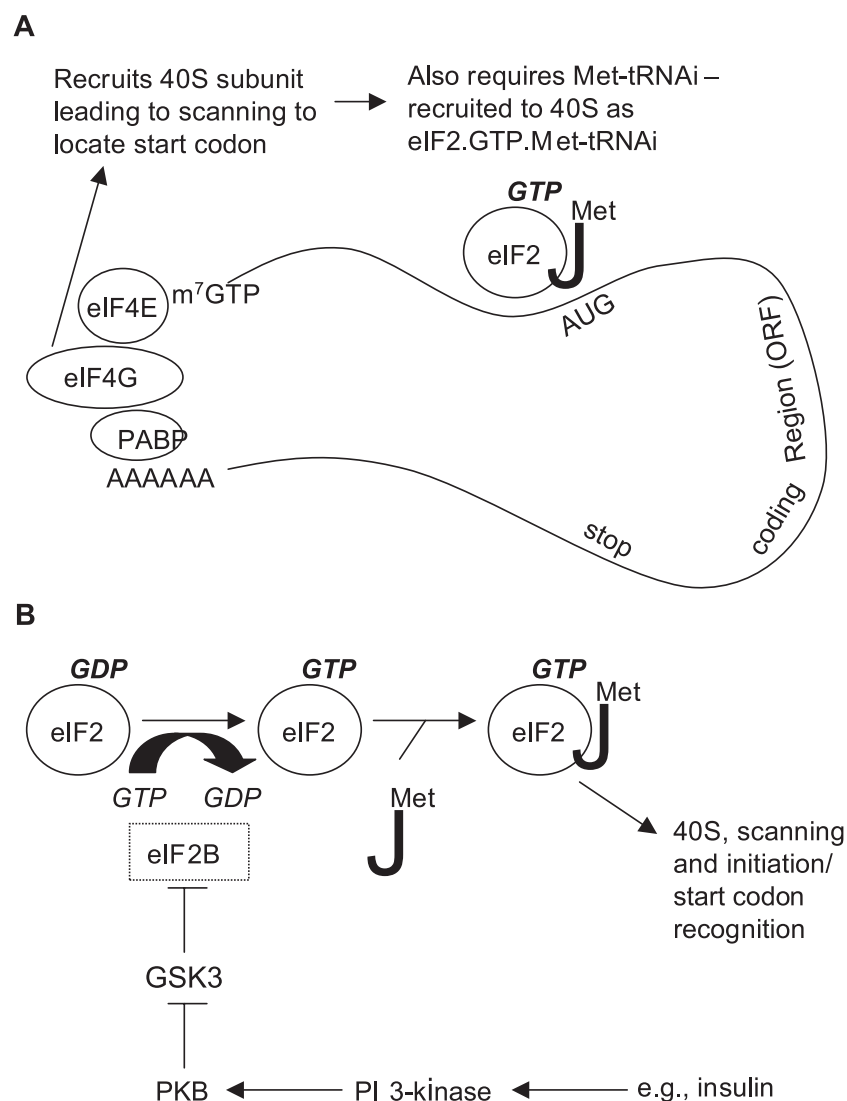


Fig. 2. mRNA translation. (A) eIF4E binds the 7-methylGTP ‘cap’ structure at the 5'-end of the mRNA and interacts with the scaffold protein eIF4G. The ability of eIF4G to bind eIF4E and PABP effectively circularises the mRNA. eIF4G also binds eIF3 (not shown) and this recruits the 40S ribosomal subunit to the mRNA. In its GTP-bound state, eIF2 binds Met-tRNA_i (see also (B)); this tRNA is shown as a large ‘J’ and this ternary complex binds the 40S subunit. The start codon is located by the anticodon on the Met-tRNA_i. Following start codon recognition, the GTP is hydrolysed and eIF2 leaves the ribosome as inactive eIF2.GDP. (B) eIF2.GDP is converted to active eIF2.GTP by guanine nucleotide exchange mediated by eIF2B. Phosphorylation of eIF2B by GSK3 inhibits its activity. GSK3 is inhibited by phosphorylation by PKB which is activated through PI3K signalling (Fig. 1C).

the cap and to the scaffold protein eIF4G. eIF4G in turn interacts with several other proteins including eIF3 (which binds to the 40S ribosomal subunit), eIF4A (an ATP-dependent RNA helicase), the poly(A)-binding protein (PABP) and the Mnks, protein kinases that phosphorylate eIF4E [63]. Binding of eIF4E to eIF4G is regulated by small heat-stable phosphoproteins called eIF4E-binding proteins (4E-BPs), as discussed below.

The 40S subunit thus forms a complex with a set of proteins that scans along the 5'-leader region of the mRNA to locate the start codon, which is recognised by the anticodon of the initiator methionyl tRNA (Met-tRNA_i, Fig. 2A). It is brought to the 40S subunit by eIF2 complexed with GTP. Following recognition of the start codon, the GTP is hydrolysed and eIF2/GDP and other factors are released from the 40S subunit. After the 60S subunit joins, the 80S ribosome proceeds into elongation. GTP/GDP exchange is needed to regenerate active eIF2.GTP after each cycle of initiation. This is mediated by eIF2B, a heteropentameric exchange factor that plays a key role in regulating translation initiation in eukaryotes from yeast to humans [29].

Elongation requires two main elongation factors in mammals, eEF1 α and eEF2 [59]. eEF1 α brings each amino acyl-tRNA to the ribosome, while eEF2 is required for the translocation process in which the ribosome moves relative to the mRNA. Both bind GTP. eEF2 can be regulated by phosphorylation, as discussed below. Encountering the stop codon triggers termination leading to release of the completed polypeptide chain and of the ribosomal subunits.

5. Control of translation

The regulation of initiation has been extensively studied, but elongation is also subject to acute control. Such control is generally exerted through changes in the phosphorylation states of the translation initiation or elongation factors that mediate these stages of translation. Phosphorylation may, for example, alter their intrinsic activity or their interactions with other components. Translation factors can be regulated through several signaling pathways and what follows focuses on those aspects that are most relevant to the topic in hand, i.e., regulation of translation through Ras/Erk and PI3K signaling.

6. Protein synthesis and hypertrophy

The pro-hypertrophic agents PE and ET-1 both activate protein synthesis in cardiomyocytes (see, e.g., Refs. [7,25,99]). As described above, increasing evidence points to a role for signaling through Ras and MAP kinases, especially the MEK/Erk pathway, in these effects [7,14,98,99,106,108]. However, the connections between the MEK/Erk pathway and the regulation of protein synthesis

are poorly understood, and delineating these links is crucial for a fuller understanding of the molecular mechanisms that underlie cardiac hypertrophy.

7. eIF2B

eIF2B comprises five different subunits of which eIF2B ϵ is the largest, possesses catalytic activity and is phosphorylated *in vivo*. Phosphorylation by glycogen synthase kinase-3 (GSK3) inhibits eIF2B activity. GSK3 is regulated by phosphorylation by PKB and is thus linked to insulin/PI3K signaling. Since phosphorylation of GSK3 by PKB leads to inactivation of GSK3 (against substrates such as eIF2B ϵ), this series of events allows eIF2B to be activated (Fig. 2B) [104] (but see also Ref. [102]). Since GSK3 can also be phosphorylated and inactivated by other protein kinases (p90^{RSK} and the p70 S6 kinase which is discussed below [85]), other signaling pathways may also feed into this regulatory mechanism (Fig. 1C). Expression of a non-phosphorylatable mutant of GSK3 β (Ser9Ala) inhibits hypertrophic responses to ET-1 *in vitro* [26] or to the β -adrenergic agonist isoproterenol or pressure overload *in vivo* [2].

A further target for PI3K signaling is the protein mTOR (mammalian target of rapamycin), which is involved in controlling a number of proteins that regulate mRNA translation [23,63]. These include the ribosomal protein S6 kinases (S6K1 and S6K2 [3]), the translational repressor proteins 4E-BP1/2 and the kinase that phosphorylates elongation factor 2 (eEF2 kinase). All these effects contribute to the activation of protein synthesis or to increases in the cellular capacity for protein synthesis (e.g., ribosome number) [3,23,51,63]. mTOR appears to coordinate inputs from hormones, mitogens and other stimuli, on the one hand, and nutrients such as branched-chain amino acids, on the other, to control protein synthesis, gene expression and cell size and cell proliferation [24,60,67,90]. Several studies point to a role for signaling through mTOR in cardiac hypertrophy: rapamycin blocks the increase in heart weight in an overload model of hypertrophy [80] and the increases in cardiomyocyte size induced by agents such as PE [7], Ang II [69] and H₂O₂ [96]. Rapamycin also blocks the activation of protein synthesis by PE or ET-1 [7,99].

8. The S6 kinases

These two kinases are products of distinct genes but have very similar amino acid sequences and in general their regulation is also similar (but not identical, see Ref. [45]). Both are stimulated by insulin and a variety of other agents and, in all cases, tested this is blocked by rapamycin. Activation of the S6Ks involves their phosphorylation at multiple sites [3]. Rapamycin prevents phosphorylation of

many sites, some of which are directly phosphorylated by mTOR, at least in vitro.

Both S6Ks phosphorylate ribosomal protein S6 at multiple sites in its C-terminus. What is the function of these enzymes? A widely held model implicates them in the upregulation of the translation of the 5'-terminal oligopyrimidine (TOP) mRNAs. This subset of mRNAs includes those for the cytoplasmic ribosomal proteins and other components of the translational machinery such as elongation factors [51]. The 5'-TOP ensures these mRNAs are translated poorly in serum-starved cells. Following serum stimulation, they are recruited into polyribosomes and translated efficiently. This effect is impaired by rapamycin and inhibition is alleviated by a rapamycin-resistant mutant of S6K1 [35], implying a role for S6K1 and perhaps S6 phosphorylation in their control. However, recent data cast doubt on a role for S6Ks in the control of 5'-TOP mRNA translation, at least in response to amino acids (which also promote their translation [87]). Control of 5'-TOP mRNA translation is likely to be important in hypertrophic responses that are associated with elevated rates of protein synthesis. Upregulation of ribosome protein synthesis (and thus ribosome biogenesis), as well as of elongation factors, will lead to an increase in the cellular capacity for protein synthesis. This will favor cell growth, i.e., hypertrophy.

Indeed, animals in which genes for S6Ks have been knocked out show reduced growth and a smaller adult size [54,77]. In *Drosophila*, the size of cells of adult S6K^{-/-} animals is clearly decreased compared to wildtype controls, although total cell numbers appear very similar. This is thus a mirror image of the situation in cardiac hypertrophy, implying that activation of the S6Ks may play a key positive role in the events that contribute to heart growth. Such stimuli include PE and ET-1 which in adult cells which activate S6K1 and 2. Activation is blocked by inhibitors of MEK and expression of activated MEK can drive phosphorylation and activation of S6K1 and 2 in cardiomyocytes [98,99]. Conversely, a dominant interfering mutant of Ras (N17Ras) blocks the activation of S6K1 by PE. This suggests that Ras/MEK signaling is important in the activation of S6Ks by these GPCR agonists. These agents do not activate PKB, and the ability of PE to activate S6K1 is not impaired by the PI3K inhibitor wortmannin, consistent with the notion that they signal to mTOR/S6Ks by different mechanisms from those employed by other more widely studied stimuli such as insulin/IGF1. It is thus important to note that rapamycin blocks cardiac hypertrophy induced by overload [80] or α_1 -adrenergic stimulation [7].

9. 4E-BPs

The 4E-BPs bind to eIF4E (thus 'eIF4E-binding protein') and prevent it from binding to other initiation factors (specifically eIF4G) to form active initiation complexes. Since eIF4E is the protein that binds the 5'-cap structure at

the start of the mRNA, the 4E-BPs act to block normal cap-dependent translation [63]. By far the best understood 4E-BP is 4E-BP1. In response to agents that stimulate mTOR signaling, e.g., PE/ET-1 in heart cells, 4E-BP1 undergoes phosphorylation leading to its release from eIF4E, which can then form functional initiation complexes with eIF4G. mTOR signaling can thus serve to stimulate general translation initiation. Again, this is likely to play a role in hypertrophic responses.

In adult cardiomyocytes, the effects of PE and ET-1 on 4E-BP1 and its binding to eIF4E are blocked by inhibition of MEK as well as by rapamycin but not by inhibiting PI3K/PKB signaling using wortmannin [99]. This again points to connections between the MEK/Erk pathway and the regulation of mTOR, an issue that is pursued in further detail below. Other evidence points to connections between MEK/Erk and mTOR signaling that are independent of PKB activation [27,28,88].

10. eEF2

Phosphorylation of eEF2 at Thr56 prevents it binding to the ribosome and thus inactivates it (reviewed in [8]). Phosphorylation is catalysed by a specific and unusual enzyme, eEF2 kinase, which is not a member of the main Ser/Thr/Tyr kinase superfamily [68], and is normally dependent on calcium ions/calmodulin for activity. Evidence has been presented for a specific cardiac isoform of eEF2 kinase [48]. Many agents that activate protein synthesis induce dephosphorylation and activation of eEF2: e.g., in cardiomyocytes, insulin, Ang II and PE and ET-1 [18,100,101]. This effect will serve to accelerate translation elongation thus contributing to activation of protein synthesis. These stimuli lead to inactivation of eEF2 kinase, and this effect is mediated by signaling through mTOR and PI3K (insulin [101]) or MEK (PE and ET-1 [100]). The latter again points to connections between MEK/(Erk) signaling and mTOR. Several mTOR-regulated phosphorylation sites exist within eEF2 kinase including Ser365 which is a target for phosphorylation by S6K1 and p90^{RSK} [103].

11. Regulation of mTOR signaling

A major question is how upstream signaling components such as PI3K/PKB lead to activation of mTOR signaling and there have recently been important advances here, stemming both from genetic studies (in man and *Drosophila*) and biochemical investigations. The tuberous sclerosis genes *TSC1* and *TSC2* encode proteins of about 130 and 200 kDa, respectively (also called hamartin and tuberin) that heterodimerize and are important regulators of mTOR (see, e.g., Refs. [44,49]). Mutations in *TSC1* or *TSC2* give rise to benign hamartomatous tumors in various tissues including the heart. Genetic studies reveal *TSC1* and *TSC2* to be

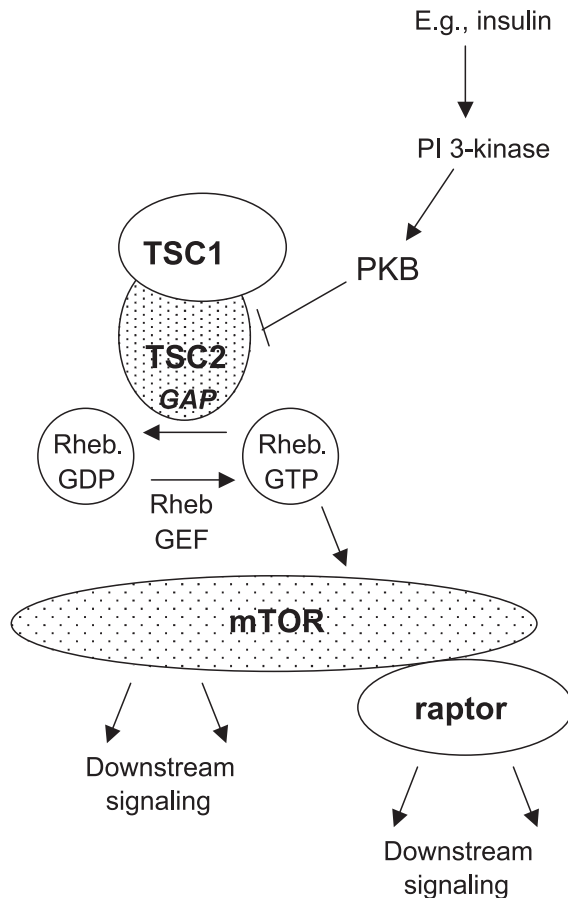


Fig. 3. mTOR signaling: TSC1/2 and Rheb. The GAP domain of TSC2 promotes hydrolysis of GTP on Rheb. Only active Rheb.GTP appears to stimulate mTOR signaling. Phosphorylation of TSC2 by PKB impairs its ability to function as a Rheb.GAP, although the mechanism involved remains unclear. PKB is activated by insulin via PI3K, thus providing a mechanism by which insulin stimulates mTOR signaling.

negative regulators of the cell cycle and cell size [33,56,58] and epistasis places them upstream of S6K but downstream of PI3K and PKB [21,58]. mTOR signaling is constitutively activated in cells from tuberous sclerosis patients. When we recall the role of mTOR signaling and S6Ks in cell size control, it is clear that activation of mTOR signaling is likely an underlying cause of the large size of these cells.

Overexpression of TSC1/2 impairs the regulation of S6K and 4E-BP1 via PI3K/PKB [31,34,89]. TSC2 is phosphorylated at at least two sites by PKB (Ser939 and Thr1462) and its phosphorylation apparently alleviates the inhibitory effect of TSC1/2 on mTOR signaling (Fig. 3), although the mechanism underlying this remains the subject of controversy.

TSC2 contains a domain with similarity to certain GTPase-activator proteins (GAPs) and recent data indicate that the molecular target of the TSC1/2 complex is the small G-protein Rheb (reviewed in [43]). TSC2 acts as a Rheb-GAP and its activity is enhanced by association with TSC1. These and other data imply that Rheb-GTP functions to activate mTOR signaling (summarised in Fig. 3). Further

work will no doubt cast further light on this issue and other aspects of the upstream control of mTOR (e.g., the key outstanding question of how amino acids feed into the regulation of mTOR).

As described above, the available data suggest that GPCRs signal via the MEK/Erk pathway to control translational components that lie downstream of mTOR. This prompts two key questions. Is regulation of mTOR by MEK/Erk signaling also mediated via TSC1/2? If so, how might PE or ET-1 activated MEK/Erk signaling feed into this control mechanism? Tee et al. [88] have shown that the ability of the MEK/Erk pathway to enhance phosphorylation

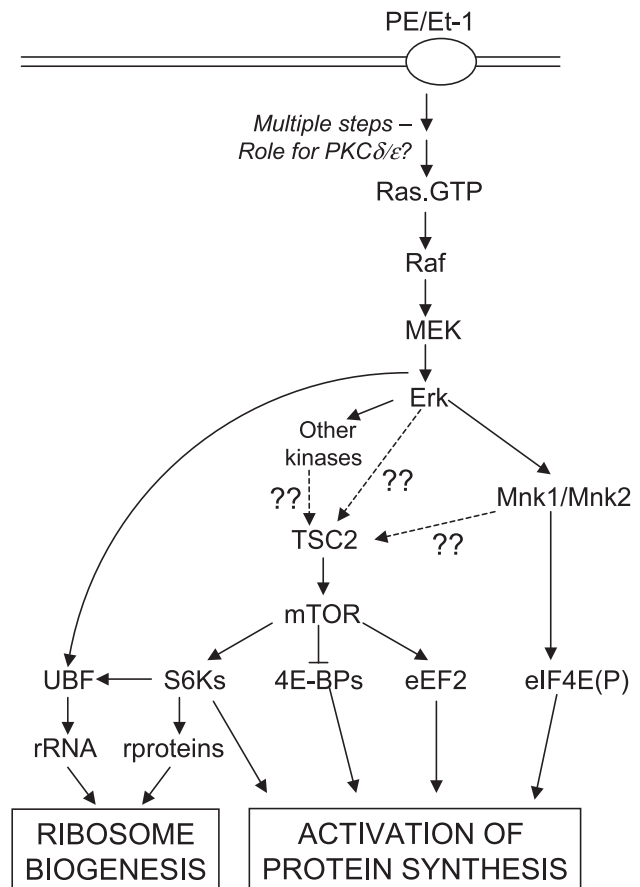


Fig. 4. Signaling connections between MEK/Erk and control of protein synthesis. Agents such as PE and ET-1 activate the Ras/Raf/MEK/Erk pathway through events that may involve PKCs δ/ϵ . Erk activates the eIF4E kinases Mnk1 and Mnk2, which phosphorylate eIF4E. This may contribute to the stimulation of protein synthesis. MEK/Erk are also required for activation by these agents of mTOR signaling, which in turn activates several components of the translational machinery, as indicated (see text for details). Links between MEK/Erk and control of mTOR remain unclear. The figure depicts the possibility that TSC2 is phosphorylated by Erk or downstream kinases, such as RSKs, MSKs or Mnk (broken arrows). Such phosphorylation, like that catalysed by PKB, may impair its GAP function towards Rheb.GTP, resulting in accumulation of active Rheb.GTP and stimulating mTOR. MEK/Erk signaling thereby contributes to overall activation of protein synthesis and ribosome biogenesis through S6Ks and enhanced translation of 5'-TOP mRNAs. Other links to ribosome biogenesis include control of UBF (and rDNA transcription) via S6Ks and Erk.

of S6K1 and 4E-BP1 is blocked by TSC1/2. This is at least consistent with the notion that control of mTOR by this pathway involves TSC1/2. Furthermore, these authors also showed that activation of the MEK/Erk pathway leads to phosphorylation of TSC2, and that this is blocked by an inhibitor of MEK. Their evidence suggests that this may involve phosphorylation of the PKB site at Thr1462 (but not mediated by PKB) and at additional sites that are not phosphorylated in response to stimulation of the PI3K/PKB pathway. A key question is how MEK/Erk brings about TSC2 phosphorylation and, in particular, the identity of the kinases involved. Since the specificity of p90^{RSK}, which lies downstream of Erk (Fig. 1A), overlaps with that of PKB, it would be a potential candidate for acting at Thr1462, although other evidence argues against this [89]. Other enzymes downstream of Erk (Mnks, Msks or RSK2-4) may be involved instead.

12. Other relevant signaling connections to the translational machinery

Erk signaling also leads to the activation of the MAP kinase-signal integrating kinases Mnk1 and Mnk2 (the latter has a much higher basal activity than Mnk1 [73,74]). The Mnks are also phosphorylated and activated by p38 MAP kinase α/β . The best understood substrate for these kinases is eIF4E, the mRNA cap-binding protein. Phosphorylation of eIF4E affects its affinity for capped mRNA, but both increased [52] and decreased [75] affinity have been reported. The role of phosphorylation of eIF4E, and thus of the activation of the Mnks by Erk signaling, in the control of protein synthesis remains unclear [74]. Nonetheless, a recent study has reported that activation of Mnks plays a key role in the hypertrophic response in rat vascular smooth muscle cells [32]. Given that both Erk and p38 MAP kinase α/β are implicated in hypertrophic responses in cardiac myocytes [19,53,83], it will be important to establish whether the Mnks, and eIF4E phosphorylation, play a role in this, especially in the activation of protein synthesis.

It is also important to note that upstream binding factor (UBF), which plays a key role in the regulation of ribosomal RNA (rRNA) gene transcription, is activated by Erk1/2 [82]. UBF is required for the transcription of rRNA genes by RNA polymerase 1. Given the recent data showing that MEK/Erk signaling mediates the activation of S6K1/2 in response to hypertrophic agents in cardiomyocytes [98,99], it appears that the MEK/Erk pathway may serve to upregulate both the synthesis of ribosomal RNA (via UBF) and the translational upregulation of ribosomal protein production (through activation of S6K and facilitation of 5'-TOP mRNA translation). Indeed, very recent data suggest that S6K1 is required for rDNA transcription through a mechanism that also involves phosphorylation of UBF [24].

13. Putting the network together

Hypertrophic agents such as PE and ET-1 appear to activate protein synthesis in adult cardiomyocytes through signaling that involves Ras, MEK and mTOR (although since rapamycin does not fully block activation [99], other mTOR-independent events may also be involved). Fig. 4 depicts signaling connections between the Ras/Raf/MEK/Erk module and regulation of components of the translational machinery. These likely include control of TSC2 through Erk or kinases that lie downstream of Erk. Further work using both biochemical approaches and transgenic animals will be required to elucidate these signaling links and to study their importance for hypertrophy in vivo.

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